

PRINCIPLES OF OPERATING SYSTEMS

S.Y.B.Sc. COMPUTER SCIENCE SEM III NEP 2020

Principles of Operating Systems

Practical No. 6

Aim: CPU Scheduling Algorithms (Part 2) – Round Robin

- Implement Round Robin scheduling with configurable time quantum.
- Compare with FCFS: fairness, turnaround, response time.
- Track context switches and improve queue management.

4. Consider the following table of arrival time and burst time for three processes P1, P2 and P3 and given Time Quantum = 2 ms. Calculate average turn around time and average waiting time. For the same set of processes, schedule them using FCFS.

Process	Arrival Time	Burst Time
P1	0 ms	5 ms
P2	4 ms	2 ms
P3	5 ms	4 ms

CODE:

```
processes = [  
    ["P1", 0, 5],  
    ["P2", 4, 2],  
    ["P3", 5, 4]  
]
```

```
quantum = 2
```

```
# ROUND ROBIN
```

```
remaining = [p[2] for p in processes]
```

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```
completion = [0, 0, 0]
queue = []
time = 0
done = 0
```

```
while done < 3:
```

```
    # Add arrived processes
```

```
    for i in range(3):
```

```
        if processes[i][1] <= time and remaining[i] > 0 and i not in queue:
            queue.append(i)
```

```
    if not queue:
```

```
        time += 1
```

```
        continue
```

```
    i = queue.pop(0)
```

```
    run = min(quantum, remaining[i])
```

```
    time += run
```

```
    remaining[i] -= run
```

```
    # Add newly arrived processes
```

```
    for j in range(3):
```

```
        if processes[j][1] <= time and remaining[j] > 0 and j not in queue and j !=
```

```
        i:
```

```
            queue.append(j)
```

```
    if remaining[i] > 0:
```

```
        queue.append(i)
```

```
    else:
```

```
        completion[i] = time
```

```
        done += 1
```

```
tat = []
```

```
wt = []
```

```
for i in range(3):
```

```
    tat.append(completion[i] - processes[i][1])
```

```
    wt.append(tat[i] - processes[i][2])
```

```
print("ROUND ROBIN")
```

```
print("Average Turnaround Time:", sum(tat) / 3)
```

```
print("Average Waiting Time:", sum(wt) / 3)
```

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```
# FCFS
time = 0
completion = []

for p in processes:

    if time < p[1]:
        time = p[1]

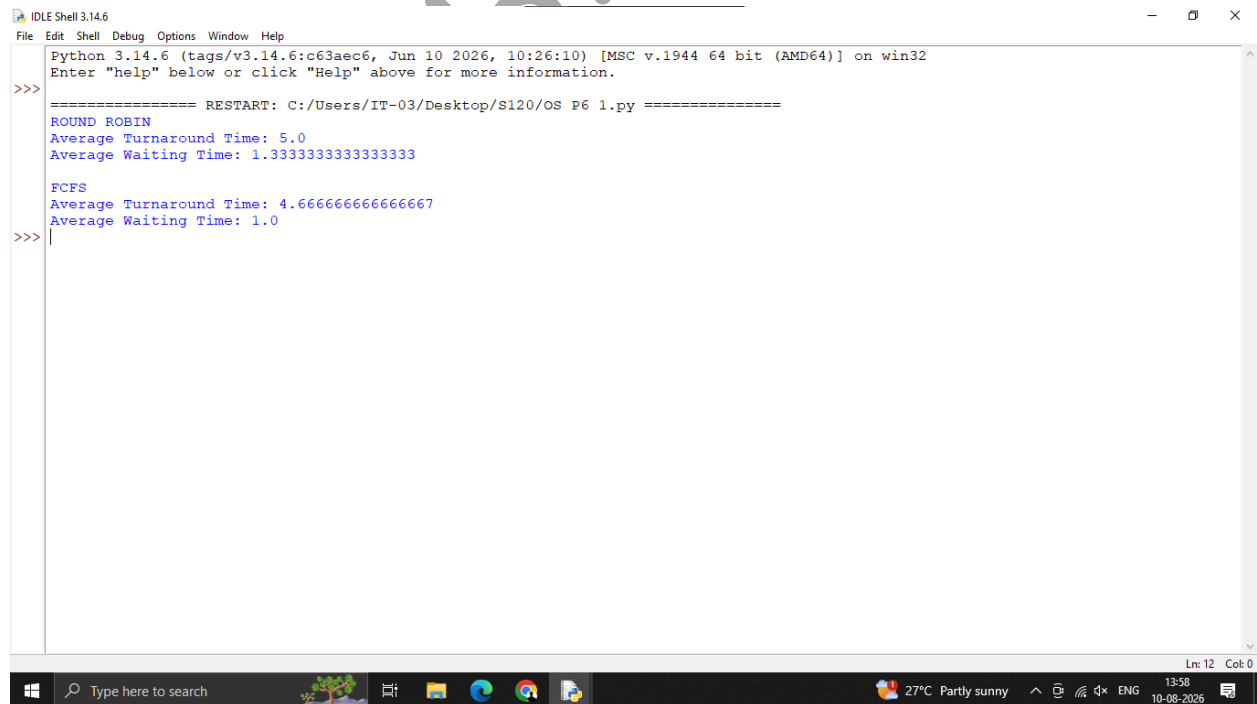
    time += p[2]
    completion.append(time)

tat = []
wt = []

for i in range(3):
    tat.append(completion[i] - processes[i][1])
    wt.append(tat[i] - processes[i][2])

print("\nFCFS")
print("Average Turnaround Time:", sum(tat) / 3)
print("Average Waiting Time:", sum(wt) / 3)
```

OUTPUT:



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: C:/Users/IT-03/Desktop/S120/OS P6 1.py =====
ROUND ROBIN
Average Turnaround Time: 5.0
Average Waiting Time: 1.3333333333333333

FCFS
Average Turnaround Time: 4.666666666666667
Average Waiting Time: 1.0
>>> |
```

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5. Consider the following table of arrival time and burst time for three processes P1, P2 and P3 and given Time Quantum = 2 ms. Calculate average turn around time and average waiting time.

Process	Arrival Time	Burst Time
P1	0 ms	5 ms
P2	1 ms	3 ms
P3	2 ms	6 ms

CODE:

```
p = [{"P1",0,5}, {"P2",1,3}, {"P3",2,6}]
q = 2
```

```
r = [5,3,6]
```

```
ct = [0,0,0]
```

```
queue = []
```

```
t = 0
```

```
done = 0
```

```
while done < 3:
```

```
    for i in range(3):
```

```
        if p[i][1] <= t and r[i] > 0 and i not in queue:
            queue.append(i)
```

```
    if not queue:
```

```
        t += 1
```

```
        continue
```

```
    i = queue.pop(0)
```

```
    x = min(q, r[i])
```

```
    t += x
```

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```
r[i] -= x

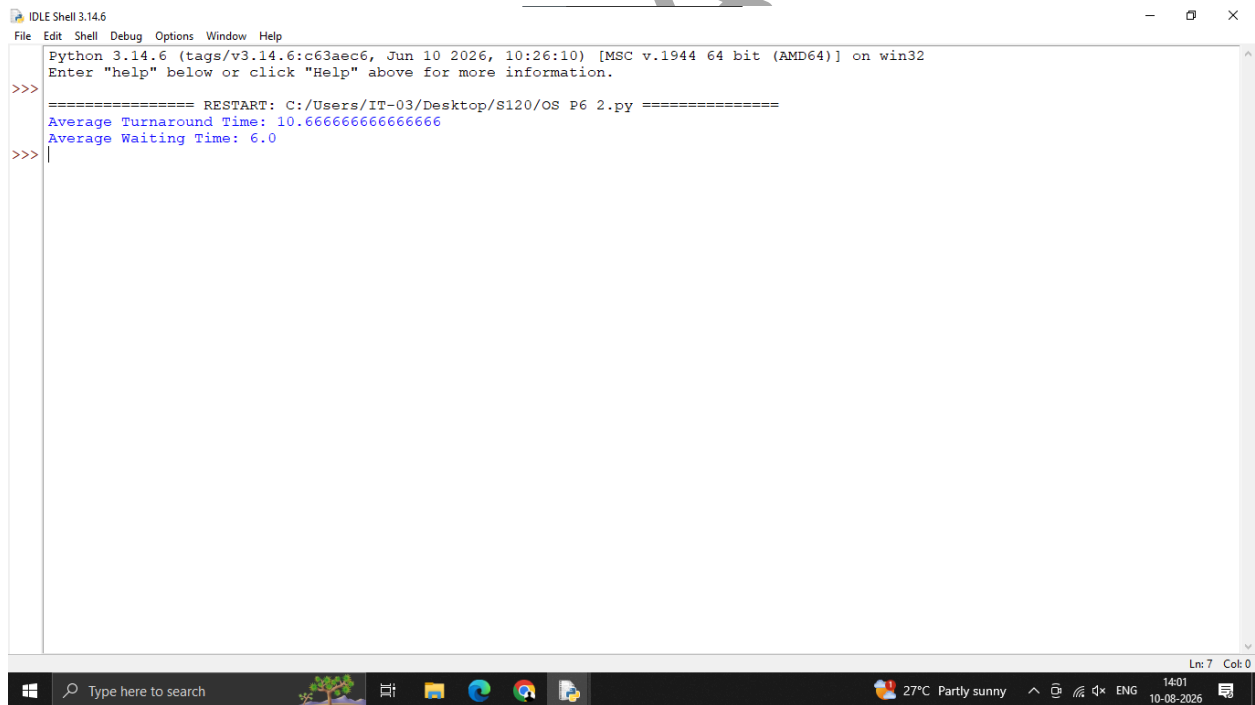
for j in range(3):
    if p[j][1] <= t and r[j] > 0 and j not in queue and j != i:
        queue.append(j)

if r[i] > 0:
    queue.append(i)
else:
    ct[i] = t
    done += 1

tat = [ct[i] - p[i][1] for i in range(3)]
wt = [tat[i] - p[i][2] for i in range(3)]

print("Average Turnaround Time:", sum(tat)/3)
print("Average Waiting Time:", sum(wt)/3)
```

OUTPUT:



```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-03/Desktop/S120/OS P6 2.py =====
Average Turnaround Time: 10.666666666666667
Average Waiting Time: 6.0
>>>
```